Selection of Materials and Design of Multilayer Lightweight Passive Thermal Protection System

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A methodology has been established aiming to design a lightweight thermal protection system (TPS), using advanced lightweight ablative materials developed at the NASA Ames Research Center. An explicit finite-difference scheme is presented for the analysis of one-dimensional transient heat transfer in a multilayer TPS. This problem is solved in two steps, in the first step, best candidate materials are selected for TPS. The selection of materials is based mainly on their thermal properties. In the second step, the geometrical dimensions are determined by using an explicit finite-difference scheme for different combinations of the selected materials, and these dimensions are optimized for the design of lightweight TPS. The best combination of material employs silicone impregnated reusable ceramic ablator (SIRCA), Saffil, and glass-wool for the first, second, and third layer, respectively. [DOI: 10.1115/1.4031737]

Keywords: thermal protection system, finite-difference method, ablation

1 Introduction

It has been demonstrated that aerodynamic heating increases with the cube of the velocity, whereas aerodynamic drag increases only as the square of the velocity [1]. Hence, at hypersonic speeds, aerodynamic heating to the vehicle increases much more rapidly with velocity than drag, and this is the primary reason why aerodynamic heating is a dominant aspect of hypersonic vehicle design. Due to very high flight speed, the aerodynamic heating increases especially during the acceleration to cruise phase [2]. It is thus mandatory to be able to predict the aerothermal environment of hypersonic vehicles in order to ensure their survivability to a high degree of certainty, without overdesigning. These basic points have forced the researchers to study about the aerothermal characteristics of the hypersonic vehicles, which is now one of the key technologies in their successful development.

TPS takes advantage of the material properties to serve as a shield for the inner structure. TPS covers the entire surface of the spacecraft's and consists of different materials on the basis of incident heat flux on the surface of the vehicle. TPS materials are used to protect the surface of the vehicle by creating a shield between aerodynamic heating load and crew compartment. In TPS design, system weight plays an important role and it includes the thickness of the TPS materials. So, it is necessary to design a lightweight TPS that could serve well under an aerodynamic heating load. On the basis of protection provided by the TPS materials, these can be categorized as: (a) ablative and (b) insulative. Ablative materials have the properties like low ablation rate, high melting point and heat capacity, and moderate thermal conductivity; due to these properties, they can withstand against high heating load. In reusable hypersonic vehicle (RHV), phenolic impregnated carbon ablator (PICA) is used to decrease the peak temperature and is used as a reusable instead of an ablator. The RHV is a fully reusable vehicle that operates at hypersonic speeds for a significant part of its trajectory. Generally, the nose caps of RHVs are sharper than the re-entry vehicles. RHVs include the military spaceplane, spaceplanes for tourism, space trucks, suborbital package delivery vehicles, and hypersonic air-breathing vehicles.

Insulative TPS material maintains reasonable temperature in inner structure, for the electronic equipment's safety. They have the properties like low density and low thermal conductivity [2]. To tackle high aerodynamic heating load, various ablative and insulative TPS have been developed [3]. In the present scenario, multilayer TPSs are used to reduce the weight. Multilayer thermal insulations have the capability to reduce the temperature up to an acceptable range. Lots of research and efforts are dedicated in the aerospace industry to study about the re-entry vehicles [4–6].

The design of hypersonic vehicles that fly at a very high Mach number creates severe challenges to the designers [5]. Aerodynamic heating is the main parameter for designing the hypersonic and supersonic vehicles. Due to very high velocity of the vehicle, a bow shock wave stands in front of the vehicles, which cause the severe heating at stagnation point. The stagnation point heat flux is also dependent on the radius of the nose cone [5]. Lots of materials have been tested in TPS design like ceramic matrix composites and ultrahigh temperature ceramics. The main disadvantage of ceramic composites is that they cannot sustain high temperatures. Hence, main emphasis is on the ceramic protection system of hafnium, zirconium, and titanium borides. These materials can sustain very high temperatures, around 1700 K at atmospheric pressure [7–9]. At present, most of the work has been dedicated to ultrahigh temperature ceramic materials and hypersonic viscous flow over sharp nose geometries [10–14].

1.1 Background and Review. Weiland et al. [15] stated that it is difficult to measure flight realistic wall temperatures and heat fluxes in wind tunnels for high velocities, since the surfaces of the model are cold. Due to difficulties in experiments and high cost, main emphasis is given to the analytical and numerical approaches to determine heat transfer to hypersonic vehicles.

Mahulikar et al. [2] revealed a temperature field history of ablative (SIRCA/PICA) and insulative (felt quartz) TPS materials, PICA and insulative TPS combination is more effective in weight reduction than SIRCA and insulative combination or ablative

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alone. Xie et al. [3] observed a transient heat transfer analysis to find out the temperature profile along the thickness direction of insulative material by using finite element method and presented optimal thickness of TPS material. Borrelli et al. [5] revealed finite element base thermal and structural response of ultrahigh temperature ceramics made nose cap. Zhu et al. [16] analyzed minimization of structural temperature insulated by functionally graded metal foam under a transient heat conduction. Ferraiuolo and Manca [17] presented a transient nonlinear one-dimensional model to evaluate the temperature field history along the thickness direction. Palmer et al. [18] described loosely coupled convective and conductive analysis for TPS sizing. Dinkelmann et al. [19] described optimization of the trajectory of an aerospace craft by simulating the unsteady heat transfer. Murray and Russell [20] presented transient thermal response for missile configuration by using an explicit coupling. Gori et al. [21] evaluated the effective thermal conductivity of an ablative material with a theoretical model and also described a transient thermal analysis to verify the temperatures of a protection structure. Martinez et al. [22,23] determined the equivalent thermal forces and stresses and also developed a method to homogenize the sandwich panel. Johnson et al. [24] performed combined deterministic and reliability optimization techniques to evaluate the optimal geometry of the stiffeners and thickness of the tank skin. Gogu et al. [25] developed a procedure to find out the best way of material selection for integrated TPS for spacecraft re-entry. Savino et al. [26] determined thermal response of ultrahigh temperature ceramics under loose and tight coupling.

1.2 Inference From Review. The heat transfer rate due to aerodynamic heating is a function of both flight trajectory and the vehicle geometry. The selection of TPS material and their thickness distribution over different regions are defined by the maximum heat flux. Highly accurate, TPS design requires TPS material temperature mapping based on coupled aerodynamic heating and heat conduction in the vehicle. Using this meticulous approach, the temperature distribution should be obtained in the thickness direction of the TPS material. Thus, the temperature distribution can be useful in the selection of TPS material and TPS sizing at various locations of the vehicle. The combination of materials also plays an important role in the design of lightweight TPS.

Generally, optimization algorithms are used to find out the best combinations of the materials that would lead to a minimal mass design. Using these comparative methods, the lightweight TPS design, through the prediction of temperature along the thickness direction and surface heat flux, is described.

1.3 Objectives and Scope. The objective of the present study is to determine the optimal thicknesses of the ablative and insulative materials and also develop a procedure to find out the best insulation materials for the design of lightweight TPS. An explicit finite-difference analysis is performed to model one-dimensional heat conduction equation combined with radiation under aerodynamic heating conditions and analyze the performance of materials and their combinations in the design of light-weight TPS.

The thermal properties like conductivity and specific heat of TPS play an important role in the comparison of materials. The main problem in the selection of a material is to satisfy both thermal and general properties (density). From the thermal point of view, a material should have low thermal conductivity and high specific heat. On the other hand, a material should have low density. For finding the best candidate materials and their combinations, lots of optimization algorithms have been discussed in previous studies. These algorithms have the potential to find out the optimal thicknesses of the materials that would lead to a minimal mass design.

The paper is organized as follows: in Sec. 2, the problem is described. In Sec. 3, the insulation materials are selected on the basis of their thermal properties by using CES EDU PACK software. In Sec. 4, the model is discretized using finite-difference scheme and the governing equations and boundary conditions of finite-difference method are presented. After that, optimal dimensions are determined for the design of lightweight TPS. Finally, results and conclusions are included.

2 Problem Description and Modeling

In this section, problem is formulated to calculate the temperature field history by using the explicit finite-difference method. A schematic of the simplified thermal problem and geometric parameters of the TPS is shown in Fig. 1. The top surface is subjected to a transient heat flux $q(t)$ and $q_r(t)$ is the radiated heat flux from the top surface. The ablative material is followed by the insulation material. The insulation materials are used to decrease the temperature at the inlet of the bottom surface. The bottom surface is assumed to be perfectly insulated.

The incident heat flux $q(t)$ was assumed to vary with time as shown in Fig. 2 [25]. This heat flux was determined with the assumption that flow is laminar [25]. The variations of heat flux are assumed to be piecewise polynomial as shown in Table 1. The instantaneous heat flux is applied before about 2200 s, where $q_{\max} = 128{,}500\ \mathrm{W/m^2}$ is the maximum intensity of the heat flux and 2200 s is the final time of the re-entry (when the vehicle has landed). The pressure differences are ignored, as they appear to be small in value. After 2200 s, only natural convection coefficient was imposed on the surface of the vehicle and its profile is shown in Fig. 2. The heat transfer in TPS is modeled as one-dimensional.

3 Material Selection for TPS

In this section, the best materials are selected for ablation and insulation. The selection is based mainly on their thermal properties. After the selection of materials, thermal analysis of the TPS is considered. In this study, SIRCA and PICA are used as an

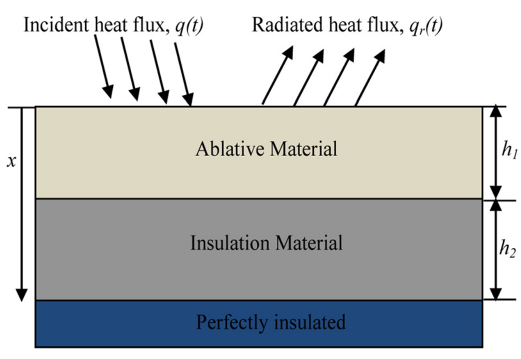

Fig. 1 A simplified model of TPS

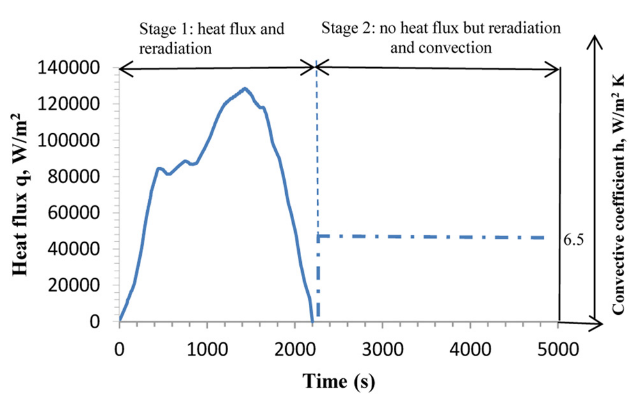

Fig. 2 Incident heat flux (stage 1) and convection coefficient (stage 2) profile with time on the TPS surface

Table 1 Polynomial fits for heat flux

Sr. no.	Time (s)	Heat flux (W/m^2)
1	0–200	$2.03 \times 10^{-3}t^3 - 42.89 \times 10^{-2}t^2 + 1440.78 \times 10^{-1}t - 35$
2	201–500	$-243.69 \times 10^{-5}t^3 + 198.66 \times 10^{-2}t^2 - 251.11 \times t + 17,312$
3	501–505	$-666.67 \times 10^{-1}t + 116,967$
4	506–521	$-1.25 \times t^2 + 125.54 \times 10^{1}t - 232,138$
5	522–527	$-666.67 \times 10^{-1}t + 117,233$
6	528–538	$-111.11 \times 10^{-2}t^2 + 114.11 \times 10^{1}t - 210,713$
7	539–737	$43.54 \times t + 56,212$
8	738–794	$-4.78 \times 10^{-1}t^2 + 714.65 \times t - 178,640$
9	795–805	$-333.33 \times 10^{-1}t + 113,733$
10	806–1000	$40.89 \times 10^{-5}t^3 - 87.49 \times 10^{-2}t^2 + 633.19 \times t - 68,955$
11	1001–1400	$-18.17 \times 10^{-2}t^2 + 507.06 \times t - 227,621$
12	1401–1800	$-2.44 \times 10^{-1}t^2 + 693.27 \times t - 364,000$
13	1801–1900	$-215.04 \times t + 480,718$
14	1901–2200	$-0.01 \times 10^{-1}t^2 - 225.34 \times t + 505,801$

ablative material because SIRCA and PICA are the advanced lightweight ablative materials developed at the NASA Ames Research Center. CES selector is used to select the insulation materials.

To compare the properties of a wide range of materials, CES EDU PACK commercial software is used, which is quite helpful to shortlist the materials [27]. This software has the material universe in which approximately 3000 materials data are present. It can also give a graphical comparison of materials, which make the selection easy. In this software, the searching of materials is allowed by imposing the constraints on their material properties. For the selection of insulation materials, following constraints are imposed:

$$\begin{aligned} &\text{min. } (T_{\text{max,service}}) > 1000\ \text{K} \\ &\text{thermal conductivity } (k) < 0.1\ \text{W/m K} \end{aligned} \tag{1}$$

The maximum service temperature must be greater than 1000 K because the transition from ablative to insulative is assumed to be 1000 K and it is decided by the maximum service temperature of the insulative material. A total of 46 materials satisfied the above constraints, the choice of materials for insulation is very high. These materials are plotted in the plane of k versus (ρ*C_p), as shown in Fig. 3. The materials are ranked according to the minimum value of thermal conductivity and low density, so Saffil, Q-fiber, carbon foam, and graphite foam are found to be the best materials compared to the rest of the materials. In Fig. 3, the

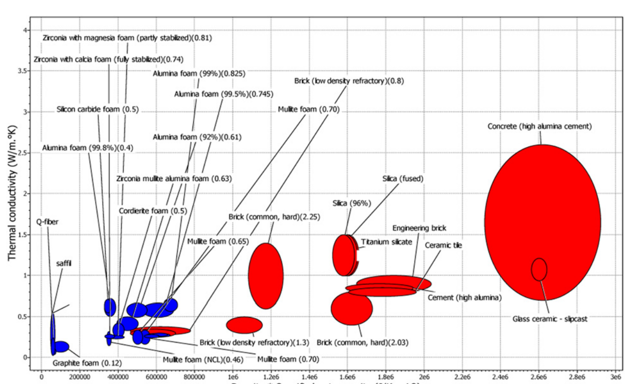

Fig. 3 First insulation layer materials, satisfying the constraints in Eq. (1)

materials are categorized in two different categories; ρ*C_p > 700,000 belongs to the ceramic class and ρ*C_p < 700,000 belongs to foams.

4 Numerical Model

The governing equation for one-dimensional transient heat conduction is given by [28]

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) = \rho C_p \frac{\partial T}{\partial t} \tag{2}$$

where ρ is the density, and k is the thermal conductivity, and C_p is the specific heat of the material, and T and t are the temperature and time, respectively.

By discretization of Eq. (2) using the explicit finite-difference method [29], the following equation is obtained:

$$\left(\frac{2k_ik_{i-1}}{k_i+k_{i-1}}\right)\frac{T_{i-1}^j - T_i^j}{(\Delta x)^2} + \left(\frac{2k_ik_{i+1}}{k_i+k_{i+1}}\right)\frac{T_{i+1}^j - T_i^j}{(\Delta x)^2} = \rho C_p \frac{T_i^{j+1} - T_i^j}{\Delta t} \tag{3}$$

where $((2k_ik_{i-1})/(k_i+k_{i-1}))$ and $((2k_ik_{i+1})/(k_i+k_{i+1}))$ denote the thermal conductivity at the middle of the two nodes $(i-1, i)$ and $(i, i+1)$, respectively [30]. The superscript j denotes the time step and the subscript i denotes the node discretization. The TPS structure is simplified into one-dimensional finite difference model as shown in Fig. 4. This model consists of three layers with different material properties. These layers are as follows: top layer, which is exposed to the surrounding and insulation and bottom layers, which are perfectly insulated. The number of nodes depends on the thickness and the node space of each layer. The space between the nodes can be different for the different layers. There are four types of nodes on the basis of their locations: exterior node, interior nodes, material interface nodes, and bottom surface node.

For the exterior node, the governing equation is given as

$$q''_{w,\text{net}} = q''_{\text{aero}} - q''_{w,\text{rad}} \tag{4}$$

where $q''_{w,\text{net}}$ is the net heat flux in the material or to the inner nodes, q''_{aero} is the aerodynamic heat flux which is applied to the exterior node, and $q''_{w,\text{rad}}$ is the radiation heat flux. By using the explicit finite-difference method, the above equation is discretized as follows:

$$T_1^j = T_1^{j-1} + \frac{2\Delta t}{\rho_1 C_{p1}\Delta x}\left[\left(q(j) + \varepsilon\sigma\left(T_s^4 - \left(T_1^{j-1}\right)^4\right)\right) + \frac{2\left(\frac{2k_1k_2}{k_1+k_2}\right)\Delta t}{\rho_1 C_{p1}\Delta x^2}\left(T_2^{j-1} - T_1^{j-1}\right)\right] \tag{5}$$

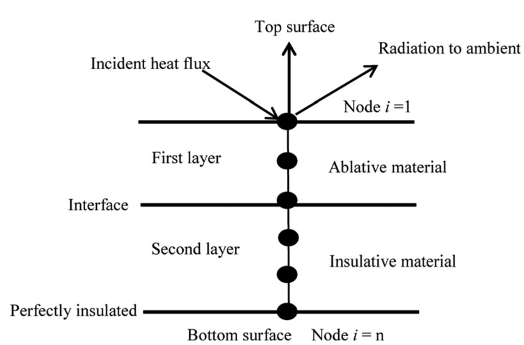

Fig. 4 Two-layer finite-difference model of TPS

where T_1 is the temperature of the exterior node and T_s is the temperature of the surrounding, assumed to be 273 K. The subscripts 1 and 2 denote the node number, ρ_1 is the density of the top surface material, k denotes the thermal conductivity of the material, C_{p1} is the specific heat, superscript j denotes the time step, Δt is the step size in the temporal coordinate, Δx denotes the step size in the spatial coordinate, $q(j)$ is the aerodynamic heat flux, and emissivity is assumed to be 0.84. Figure 5 shows the interior and material interface nodes

$$T_i^j = T_i^{j-1} + \frac{2\Delta t}{\Delta x_1 \rho_1 C_{p1} + \Delta x_2 \rho_2 C_{p2}} \left[\frac{\left(\frac{2k_i k_{i+1}}{k_i + k_{i+1}}\right)\left(T_{i+1}^{j-1} - T_i^{j-1}\right)}{\Delta x_2} - \frac{\left(\frac{2k_i k_{i-1}}{k_i + k_{i-1}}\right)\left(T_i^{j-1} - T_{i-1}^{j-1}\right)}{\Delta x_1} \right] \tag{6}$$

where subscript i denotes the interior nodes and Δx_1 and Δx_2 are the step size for layers 1 and 2. The first objective of finite-difference scheme is to find out the maximum time step (Δt) for which the solutions are stable, then check the dependency of solutions over Δt, the solutions should not vary with the Δt. The explicit finite-difference scheme is stable under the following conditions, which are given below:

$$\Delta t \leq \frac{\Delta x_1^2}{2\alpha} \tag{6}$$

$$\Delta t \leq \frac{\Delta x_2^2}{2\alpha} \tag{7}$$

where $\alpha = (k/(\rho C_p))$ denotes the maximum thermal diffusivity of the material in both the layers. The space discretization is assumed to be 1-mm, because of a better solution and the maximum time step is determined from Eq. (7). At this time step, the solutions are independent.

The end node or perfectly insulated node is subjected to zero heat flux boundary condition because the bottom face sheet is assumed to be perfectly insulated. As a result, the equation becomes

$$T_{\text{end}}^j = T_{\text{end}}^{j-1}\left(1 - \frac{2\left(\frac{2k_{\text{end}}k_{\text{end}-1}}{k_{\text{end}} + k_{\text{end}-1}}\right)\Delta t}{\rho_l C_{pl} \Delta x_l^2}\right) + \left(\frac{2\left(\left(\frac{2k_{\text{end}}k_{\text{end}-1}}{k_{\text{end}} + k_{\text{end}-1}}\right)\right)\Delta t\left(T_{\text{end}-1}^{j-1}\right)}{\rho_l C_{pl} \Delta x_l^2}\right) \tag{8}$$

where subscript "end" denotes the end node, ρ_l and C_{pl} are the density and specific heat of the end node material, respectively, and Δx_l is the step size for end node.

The geometry of the TPS is mainly defined by the two geometric variables: (1) Ablative material thickness of the first layer, h_1 and (2) insulation thickness of the second layer, h_2.

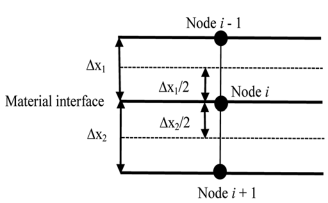

Fig. 5 Material interface node

Initially, solve for the two layers of TPS, and find out the best material combination for lightweight TPS. After that, further reduction of its weight will be considered. To find the optimal thicknesses of the insulation materials, the ranges of the thicknesses are assumed as

$$\begin{aligned} h_1 &= 2\text{–}200\,\text{mm} \\ h_2 &= 2\text{–}200\,\text{mm} \end{aligned} \tag{9}$$

The problem is formulated as

$$\begin{aligned} &\text{minimize: } m'' = \rho_1 h_1 + \rho_2 h_2 \\ &T_{1,\max} \leq 1000\,\text{K} \\ &T_{2,\max} \leq 400\,\text{K} \end{aligned} \tag{10}$$

m'' is the mass per unit area of the panel, T_1 is the transition temperature from the first layer to the second layer, and T_2 is the temperature of the bottom surface. The assumed ranges of thicknesses are taken on the basis of previous studies. The above optimization problem can be solved by using different optimization algorithm like globally convergent method of moving asymptotes (GCMMA) used by Xie et al. [3] and MATLAB (*fmincon*) function used by Zhu et al. [16]. But in this study, optimal thicknesses are determined without using any optimization algorithm.

5 Results and Discussion

In this section, thermal responses of different combinations of ablative and insulative materials under transient heating conditions are presented to demonstrate the difference in performance between the different combinations of materials and also to check the weight of the TPS. The combinations of ablative and insulative materials are shown in Table 2. The heat flux imposed on the TPS is assumed to be divided into 14 steps as shown in Table 1. The initial temperature is assumed to be 273 K and the thermal properties of the materials are given in Tables 3 and 4. In the first part of this section, PICA is used as an ablative material and in the second part PICA is replaced by SIRCA but the insulation materials remain the same.

5.1 PICA. Figure 6 shows the temperature plots for PICA (ablative) followed by Saffil (insulative). The thermophysical properties of PICA are provided in Table 4. The properties of PICA were determined at the NASA Ames [31], and the variations of the properties with temperature are shown in Table 5. Figure 6(*a*) gives radiation equilibrium temperature, Fig. 6(*b*) gives temperature profile along the thickness direction of TPS for selected time steps, and Fig. 6(*c*) gives the contact surface temperature between ablative and insulative materials and also gives the bottom surface temperature for the entire trajectory. T_1 denotes the contact surface temperature and T_2 gives the bottom surface temperature.

Table 2 Material combinations considered for analysis

Test case	Ablation material	Insulation material	Transition temperature (K)
1	PICA SIRCA	Saffil	1000
2	PICA SIRCA	Q-fiber	1000
3	PICA SIRCA	Carbon foam	1000
4	PICA SIRCA	Graphite foam	1000
5	PICA SIRCA	Glass-wool	873

Table 3 Thermophysical properties of insulative materials

Insulation materials	Density (kg/m^3)	Specific heat (J/kg K)	Max. service temp. (K)	Thermal conductivity (W/m K)
Saffil	50	942–1340	1273–1975	0.036–0.544
Q-fiber	56	841–1190	1533	0.030–0.242
Carbon foam	51	1210–1310	3650–3800	0.04–0.11
Graphite foam	70	750–800	2850–2960	0.07–0.20
Glass-wool	26	1000	873	0.03256

Table 4 Thermophysical properties (k and C in temperature range, 277–1673 K)

TPS material	k (W/m K)	C (J/kg K)	ε	ρ (kg/m^3)
PICA	0.121–0.690	1381–2093	0.84	144
SIRCA	0.02670–0.14817	670–1269	0.85	240

Table 5 Variation of $k(T)$ and $C(T)$

TPS materials	k [T(K)] (W/m K)	C [T(K)] (J/kg K)
PICA	$1.931\times10^{-10}T^3-2.287\times10^{-7}T^2+2.392\times10^{-4}T+0.0647$	$0.51\times T+1239.7$
SIRCA	$9\times10^{-5}\times T+0.0017$	$0.4291\times T+551.14$
Saffil	$2\times10^{-7}\times T^2-8\times10^{-5}T+0.0423$	—
Q-fiber	$8\times10^{-12}\times T^3+7\times10^{-9}\times T^2+5\times10^{-5}\times T+0.0161$	—
Carbon foam	$2\times10^{-5}\times T+0.0343$	—
Graphite foam	$5\times10^{-5}\times T+0.056$	—

Table 6 Optimal thicknesses for each test case with PICA

Test cases	Insulation material	Optimal thicknesses (mm)	Mass per unit area (kg/m^2)
1	PICA	20	6.58
	Saffil	74	
2	PICA	28	7.95
	Q-fiber	70	
3	PICA	32	8.28
	Carbon foam	72	
4	PICA	28	10.40
	Graphite foam	91	
5	PICA	52	9.10
	Glass-wool	62	

Radiation equilibrium is the result of the incident heat flux balanced by the emission radiation flux and gives the maximum surface temperature for a given surface emissivity and incident heat flux. The temperature at the top surface of the TPS reaches its maximum value at about 1500 s. The transition from ablative to insulative is decided by the limit of maximum service temperature of the insulation material. Once the temperature is lowered by the ablative material up to the maximum service temperature limit of insulation material, the insulative material can instantly follow to finally reduce the temperatures. The combination of ablative and insulative materials can be effective for reducing the weight of the

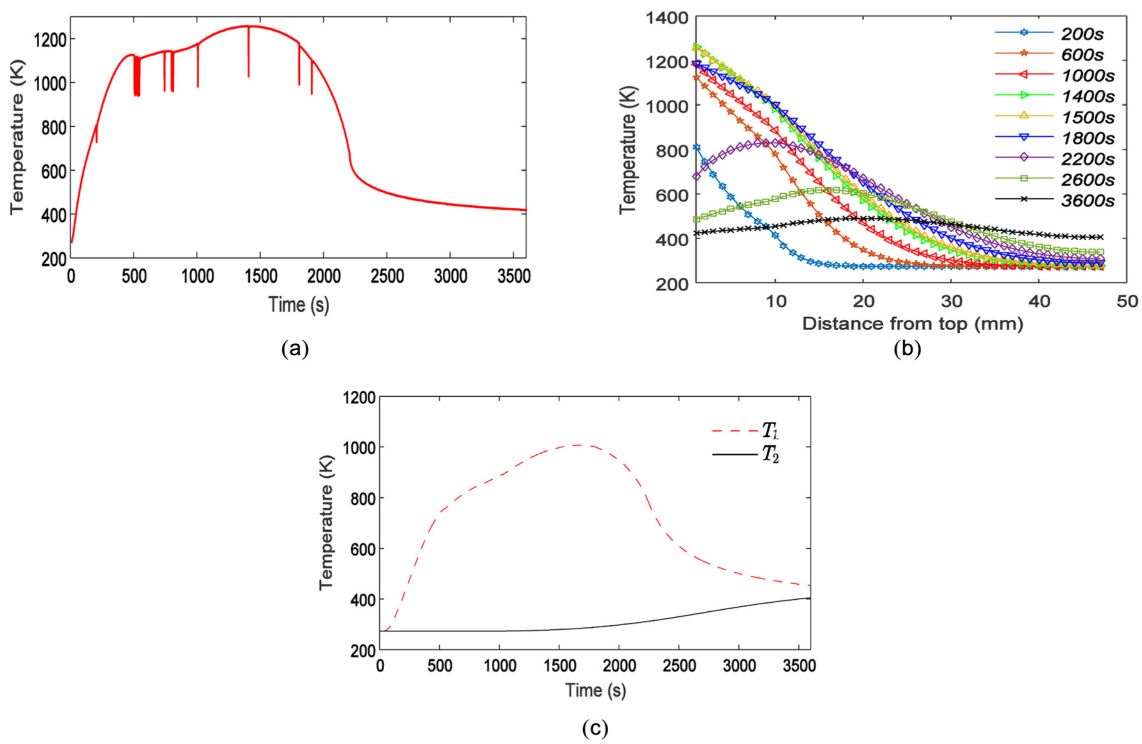

Fig. 6 (*a*) Radiation equilibrium temperature profile for PICA, (*b*) temperature profiles along the thickness direction comprising of PICA and Saffil, and (*c*) temperature profiles of contact surface temperature T_1 and bottom surface temperature T_2 comprising of PICA and Saffil

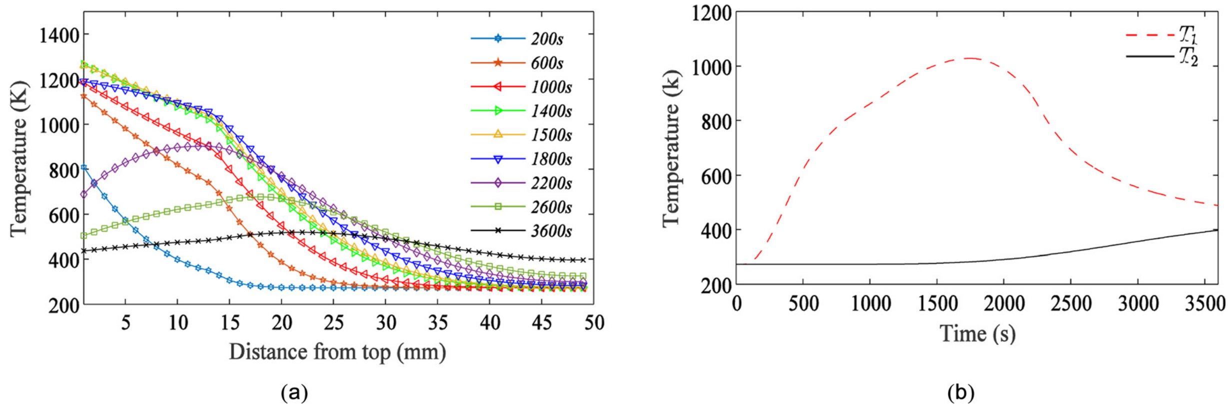

Fig. 7 (*a*) Temperature profiles along the thickness direction comprising of PICA and Q-fiber and (*b*) temperature profiles of contact surface temperature T_1 and bottom surface temperature T_2 comprising of PICA and Q-fiber

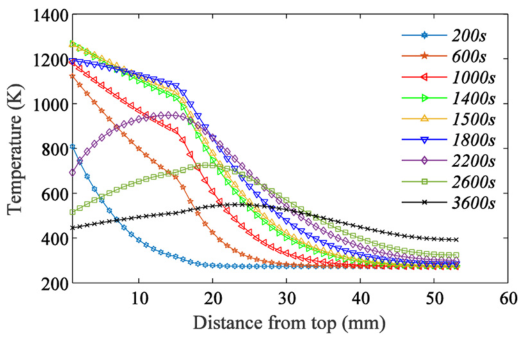

Fig. 8 Temperature profiles along the thickness direction comprising of PICA and carbon foam

TPS. This numerical model can hold for the different materials. In this analysis, the transition from ablative materials (PICA/SIRCA) to insulative materials is at the thickness location where the maximum temperature of the TPS is closest to 1000 K except for glass-wool. The transition temperatures are shown in Table 2.

The temperature at the contact surface of the ablative and insulative material reaches its maximum value at 1800 s and bottom node reaches its maximum value at 3600 s. Due to the high value of incident heat flux as shown in Fig. 2, the temperature gradient is very high but after some time (2200 s) it approaches to zero at about 3600 s. The optimal thicknesses of first and second layers of insulation are provided in Table 6.

Figure 7 gives the results for the TPS made up of PICA and Q-fiber. Figure 7(*a*) gives the temperature variations along the thickness direction of the TPS and Fig. 7(*b*) gives the contact surface and bottom surface temperatures. Temperature plots for the TPS made of carbon foam, graphite foam, and glass-wool insulative materials are given at selected time steps in Figs. 8–10, respectively. The radiation equilibrium temperature profile and contact surface temperature profiles of these cases are not presented here because they are identical to case 1. The optimal thicknesses of ablative and insulative materials and mass per unit area of the combinations are presented in Table 6.

Figures 6–10 show the variation of the temperature along the thickness direction of TPS for selected time steps. They also show the time instants, when TPS internal positions reach the peak temperature and the temperature gradient of the structure at each characteristic time point. At lower time steps (i.e., 200 s–2200 s), the protection system is heated by heat flux as shown in Fig. 2, due to this heat flux the temperature gradient is very high, but after 2200 s, only constant natural convection is imposed on the surface of the vehicle that is why the temperature profile for higher time steps (2200 s–3600 s) is different. At about 3600 s, the structure will not continue to be heated up, as the temperature gradient approaches zero.

5.2 SIRCA. In this section, PICA is replaced by low thermal conductivity material SIRCA. The thermophysical properties of SIRCA are given in Table 4. The variations of thermal conductivity and specific heat with temperature are assumed to be linear because of the lack of experimental data.

Figure 11 shows the temperature plots for SIRCA (ablative) followed by Saffil (insulative). The peak stagnation temperature in Fig 11(*a*) ($T_{\max,\mathrm{SIRCA}} = 1268$ K) is slightly higher than in Fig. 6(*a*) ($T_{\max,\mathrm{PICA}} = 1259$ K) for the same time step, i.e.,

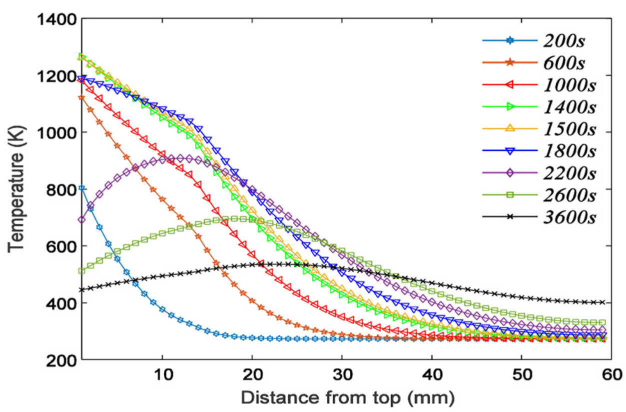

Fig. 9 Temperature profiles along the thickness direction comprising of PICA and graphite foam

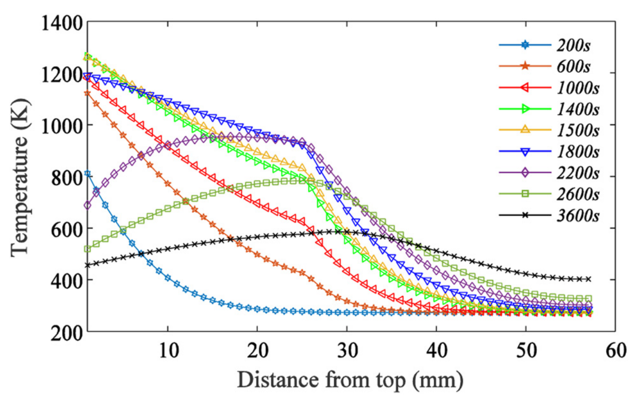

Fig. 10 Temperature profiles along the thickness direction comprising of PICA and glass-wool

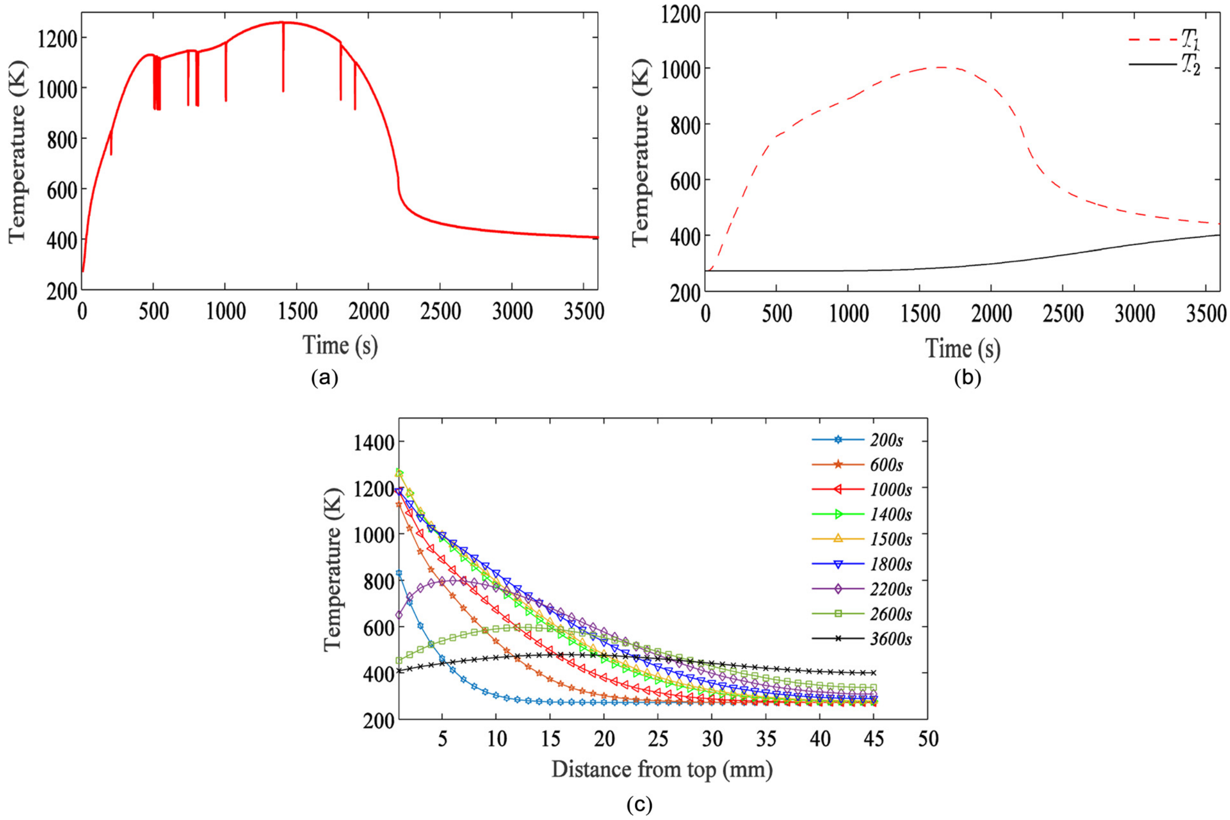

Fig. 11 (*a*) Radiation equilibrium temperature profile for SIRCA, (*b*) temperature profiles of contact surface temperature T_1 and bottom surface temperature T_2 comprising of SIRCA and Saffil, and (*c*) temperature profiles along the thickness direction comprising of SIRCA and Saffil

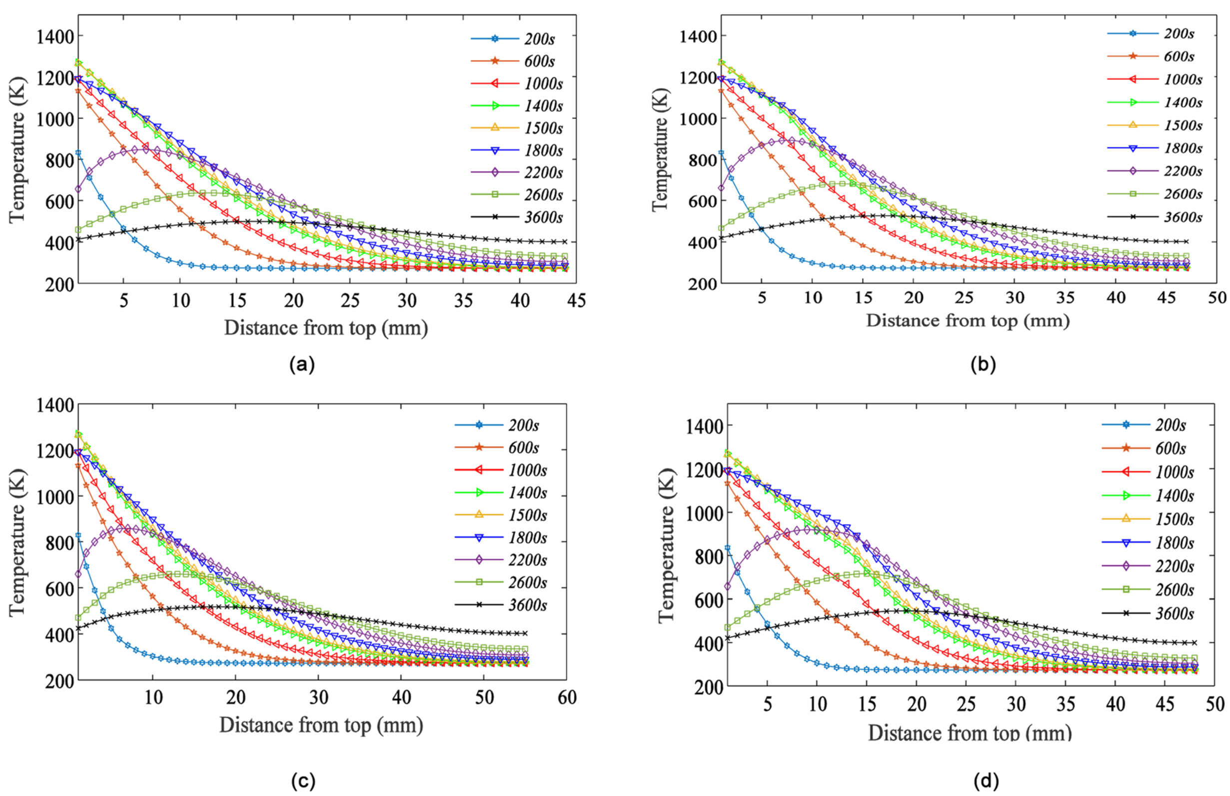

Fig. 12 Temperature profiles along the thickness direction comprising of (*a*) SIRCA and Q-fiber, (*b*) SIRCA and carbon foam, (*c*) SIRCA and graphite foam, and (*d*) SIRCA and glass-wool

Table 7 Optimal thicknesses for each test case with SIRCA

Test cases	Insulation material	Optimal thicknesses (mm)	Mass per unit area (kg/m^2)
1	SIRCA	10	6.50
	Saffil	82	
2	SIRCA	14	7.50
	Q-fiber	74	
3	SIRCA	16	7.81
	Carbon foam	78	
4	SIRCA	12	9.74
	Graphite foam	98	
5	SIRCA	28	8.48
	Glass-wool	68	
6	SIRCA	10	6.26
	Saffil	45	
	Glass-wool	62	

$T_{\max,\mathrm{SIRCA}} > T_{\max,\mathrm{PICA}}$. SIRCA has a lower thermal conductivity than PICA; thus, its peak stagnation temperature is slightly higher than the PICA. But its optimized thickness is lower than the PICA because higher conductivity material has a larger optimized thickness. Due to low thermal conductivity of SIRCA, less heat spreading takes place from the stagnation region, resulting in temperature build up in the stagnation region.

Figures 12(*a*)–12(*d*) show the temperature profile along the thickness direction of TPS for all the cases shown in Table 5. The combination of SIRCA followed by Saffil is found to be the lightest as compared to the rest of the combinations, as shown in Table 7. Comparison of Tables 6 and 7 reveals that the temperatures in the TPS are lowered as follows: (a) lower conductivity of the ablative material reduces the optimized thickness of the TPS, (b) moderate conductivity of the ablative material reduces the peak stagnation temperature, and (c) low conductivity of the insulative material reduces the temperature and optimized thickness of the insulation part of the TPS, and the results reveal that the weight of SIRCA and Saffil combination is lighter than the PICA and Saffil combination.

Finally, a three-layer model is designed for testing whether the three-layer design has the potential for further weight reduction. The weight of SIRCA and Saffil combination is lighter than the other combinations, and this combination is utilized for further improvement with respect to its weight. For that purpose, one more layer is attached after the first insulating layer (Saffil). The new design consists of SIRCA (ablative), Saffil (first layer of insulation), and glass-wool (second layer of insulation). Glass-wool is selected for second insulation layer because of its low density, as shown in Table 3. The transition from Saffil to glass-wool is at the thickness location where the maximum temperature of the TPS must be lower than 873 K. Figure 13(*a*) gives the contact surfaces temperature, T_1 denotes the contact surface temperature of SIRCA and Saffil, T_2 denotes the contact surface temperature of Saffil and glass-wool, and T_3 gives the bottom surface temperature. All the boundary conditions and initial conditions are same as previous cases. Same ranges of thicknesses have taken to determine the optimal thicknesses for the thermal insulation. Figure 13(*b*) shows the variation of temperature with the thickness of the material. This methodology can be used to find the optimal dimensions for the three layers. In addition, it gives the minimum value of mass per unit area as compared to two-layer design. The results show that the multilayer insulation is more effective as compared to the single-layer designs.

6 Summary and Conclusions

In this study, a material selection procedure is developed to find out the best candidate materials for thermal insulation. After that, a finite-difference model to discretize the one-dimensional transient heat conduction equation is developed and the optimal thicknesses for low-mass design are found out. The entire study is completed in two steps. In the first step, best candidate materials are selected, mainly on the basis of their thermal properties. For that purpose, CES EDU-PACK software is used in which 3000 materials database is available. In the second step, optimal thicknesses are determined by using a finite-difference model and materials are ranked according to their mass per unit area. Based on the results obtained, the main observations are summarized as follows:

(a) Ablative TPS material with low thermal conductivity, e.g., SIRCA, reduces the optimized thickness of ablative material relative to a material with moderate/high conductivity, e.g., PICA. But the peak stagnation temperature of low conductivity material (SIRCA) is higher than moderate conductivity material (PICA).
(b) The different material combinations are analyzed and showed that the SIRCA and Saffil lead to the design of lightweight TPS with a mass per unit area of 6.50 kg/m^2 and the combination of SIRCA and Saffil followed by glass-wool has also shown the potential for further weight reduction ($m = 6.26$ kg/m^2).
(c) The results indicated that the most effective way to reduce the optimal thicknesses of TPS is that the ablative and insulative materials must have low thermal conductivity.
(d) The procedure developed for selection of insulation materials is found very efficient and effective to obtain the best candidate materials.
(e) In this study, the optimal dimensions are determined by using the explicit finite-difference method without considering any optimization algorithm, and results indicated that this approach is very effective and accurate.

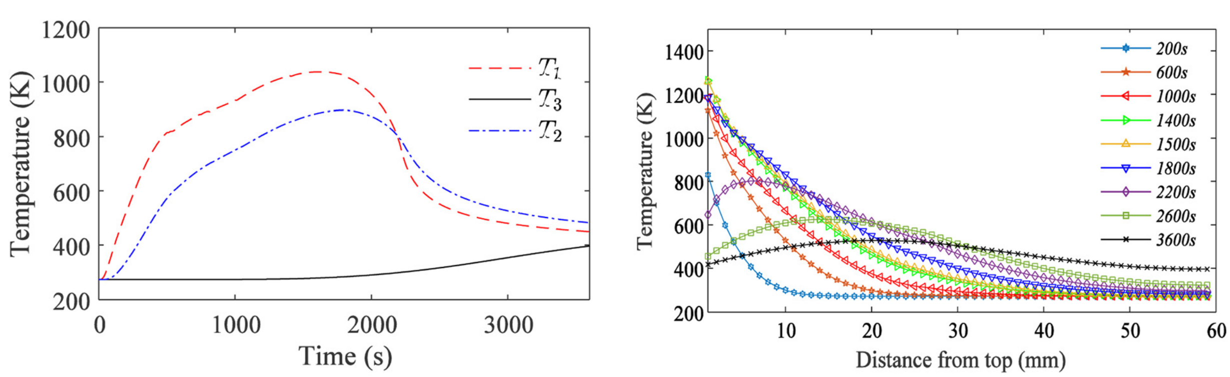

Fig. 13 (*a*) Temperature profiles of contact surface temperatures T_1, T_2, and bottom surface temperature T_3 comprising of SIRCA, Saffil, and glass-wool and (*b*) temperature profiles along the thickness direction comprising of SIRCA, Saffil, and glass-wool

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Nomenclature

C = specific heat (J/kg K)
h = material thickness (mm)
k = thermal conductivity (W/m K)
M = Mach number
P = pressure (Pa)
q = heat flux at the surface (W/m^2)
t = time (s)
T = temperature (K)

Greek Symbols

α = thermal diffusivity (m^2/s)
ε = emissivity of surface
ρ = density (kg/m^3)
σ = Stefan–Boltzmann constant ($= 5.67 \times 10^{-8}$ W/m^2 K^4)

Subscripts/Superscripts

aero. = aerodynamic
rad. = radiation
stag. = stagnation region
w = surface/wall
2D = two-dimensional
∞ = free-stream condition

Abbreviations

GCMMA = globally convergent method of moving asymptotes
PICA = phenolic impregnated carbon ablator
RHV = reusable hypersonic vehicle
SIRCA = silicone impregnated reusable ceramic ablator
TPS = thermal protection system

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